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HOW DOES A BIKE-SHARE NAVIGATE SPEEDY SUCCESS?

A COMPREHENSIVE DATA ANALYSIS OF 2025 USER BEHAVIOR

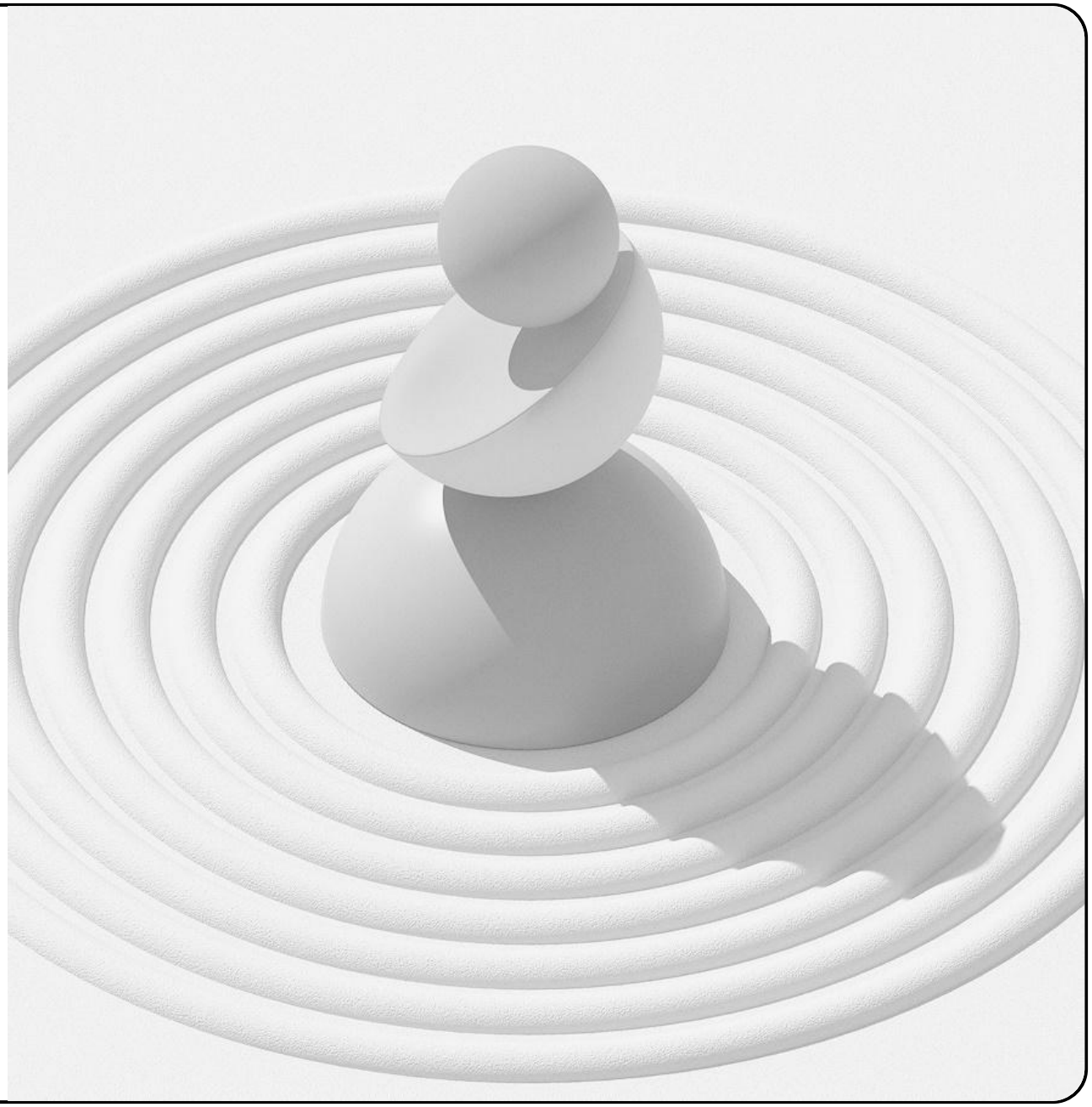
DATE: APRIL 2026

OUR OBJECTIVE

The Problem: Casual riders are a significant portion of our business, but annual members are the key to long-term, sustainable growth.

The Question: How do annual members and casual riders use Cyclistic bikes differently?

The Goal: Identify data-driven trends to design a marketing strategy that converts high-value casual riders into annual members.



Data Scope & Processing

The Dataset:

The dataset consists of Cyclistic's historical trip records covering 12 months (January 2025 – December 2025).

Volume: 5,552,994 total trips analyzed.

Data integrity: * combined 12 monthly csv files into a single integrated dataframe for a full-year view using **python/pandas**

Data is sourced directly from internal company records.

Note that for privacy, all personally identifiable information (PII) has been removed.

Documentation of Data Cleaning & Manipulation

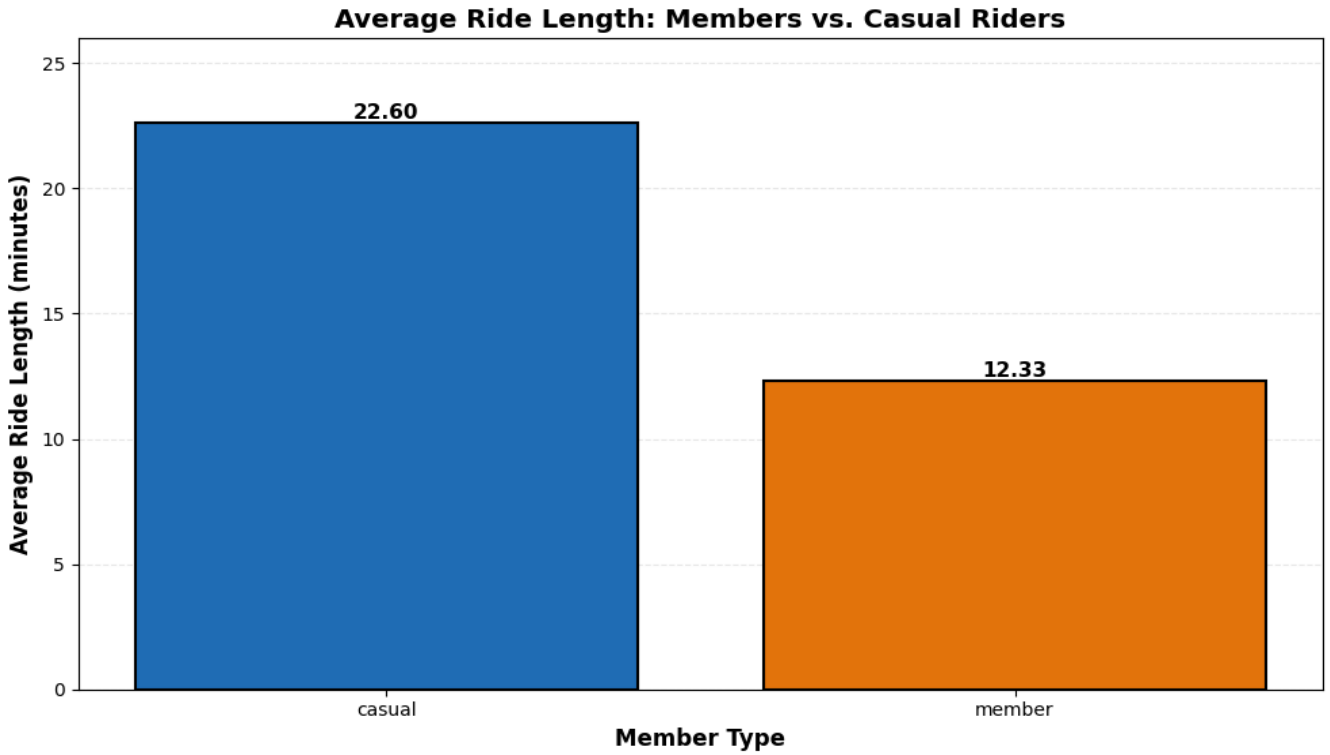
Handling Overnight Rides: Calculated duration_minutes by subtracting started_at from ended_at. Handled "midnight crossovers" by ensuring the calculation used full datetime objects.

Handling DST and Timezones: Raw timestamps produced negative ride durations during Daylight Savings Time (DST) shifts in the Chicago area. Localized all timestamps to the America/Chicago timezone.

Feature Engineering: Created new columns for duration_minutes and weekday to enable aggregation.

Calculated the duration in minutes.

Leisure vs. Utility: How Users Value Their Time



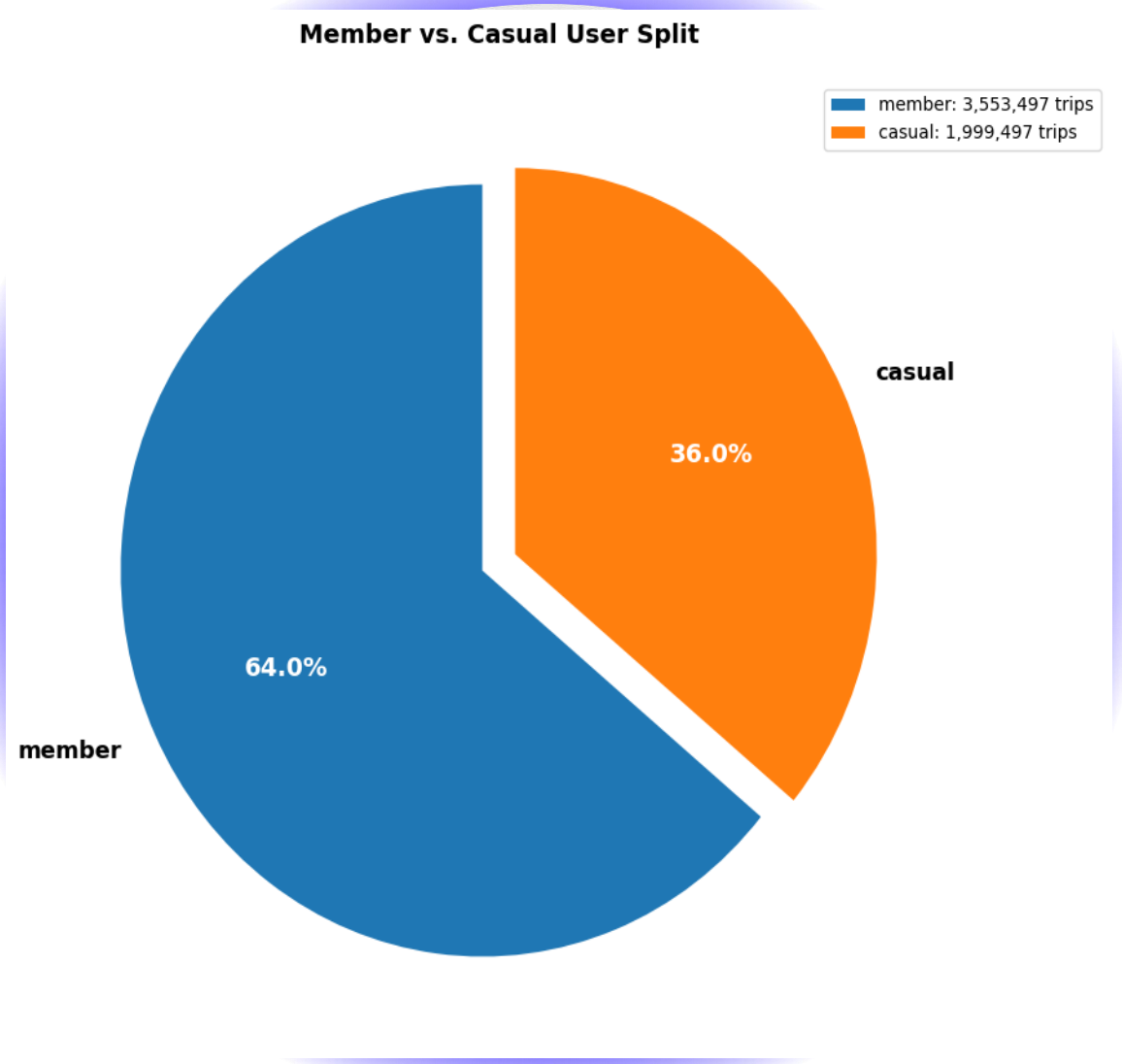
Key Findings:
Casual Riders: Average **22.60 minutes** per trip.
Annual Members: Average **12.33 minutes** per trip.
The Insight: Casual riders are not just using bikes for transportation; they are using them for *experiences*. Their trips are **83% longer**, suggesting they prioritize the journey over the destination.

USER BASE COMPOSITION

CASUAL RIDERS REPRESENT 36% OF TOTAL ANNUAL VOLUME

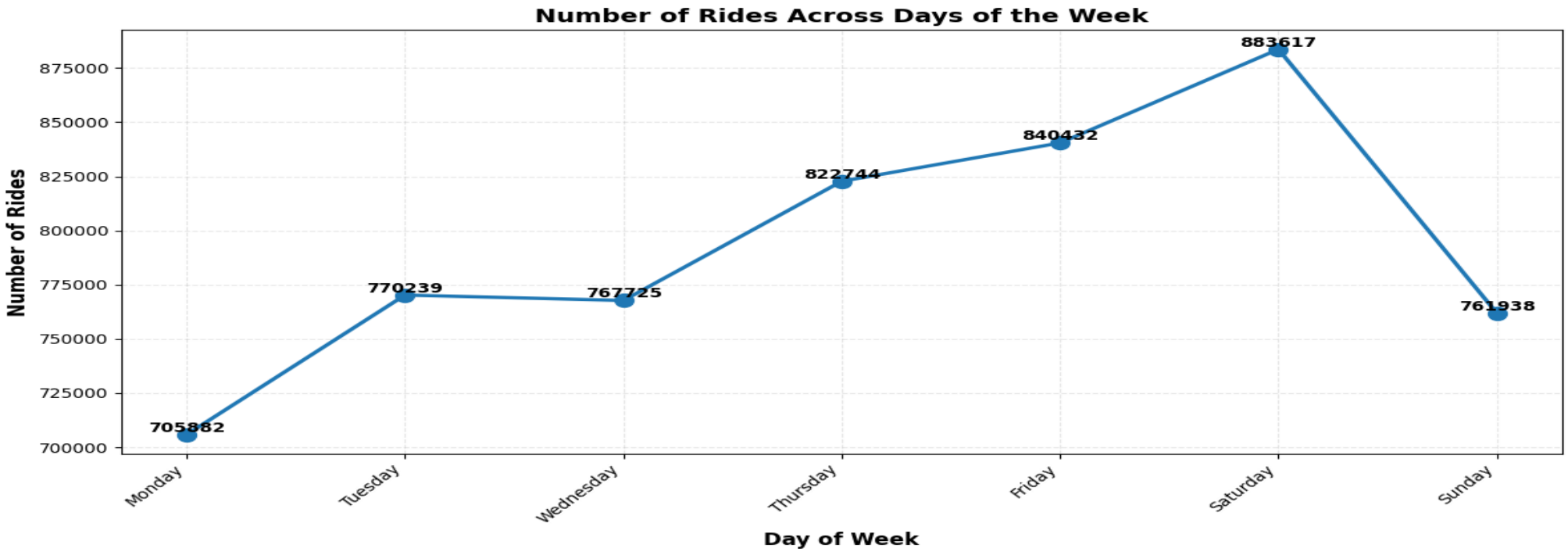
Annual Members (64%): Form the backbone of the service with over **3.5 million trips**.
Casual Riders (36%): Account for approximately **2 million trips**, representing a massive conversion opportunity.

Metric	Annual Members	Casual Riders
Total Trips	3,553,497 (64%)	1,999,497 (36%)
Avg. Trip Duration	12.33 Minutes	22.60 Minutes
Peak Usage Day	Tuesday / Thursday	Saturday
Primary Intent	Commuting / Utility	Recreation / Leisure



WEEKLY TRENDS

PEAK ENGAGEMENT OCCURS ON SATURDAYS



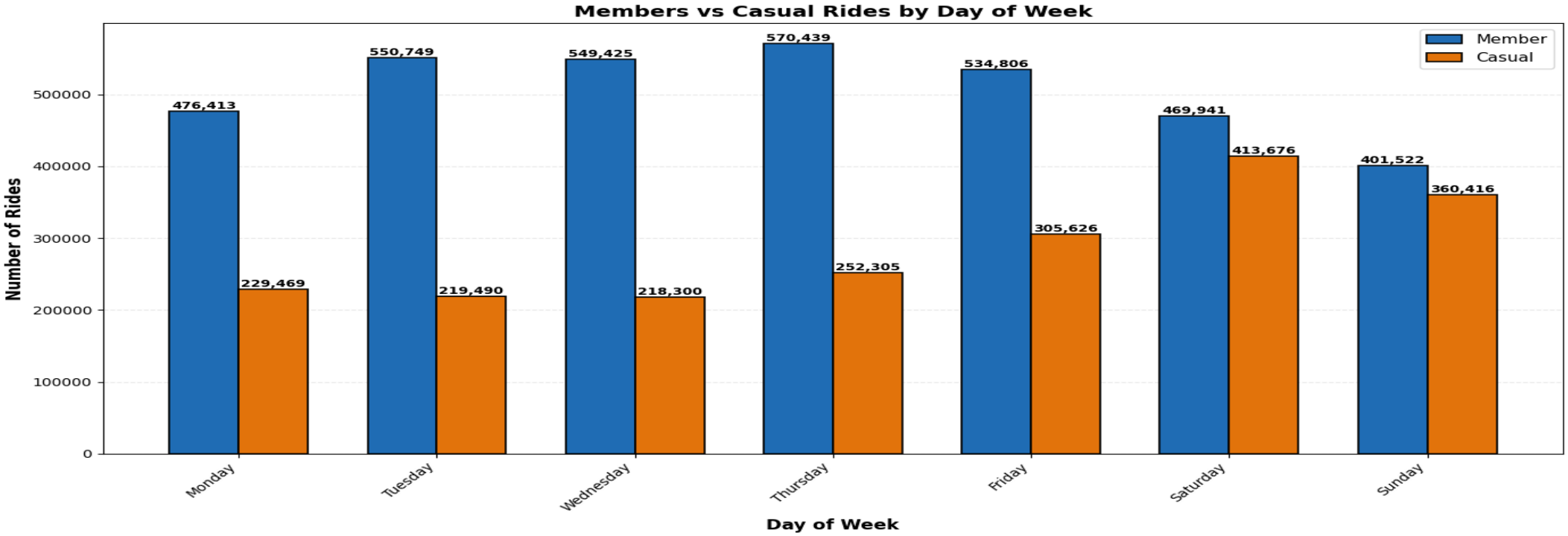
KEY FINDINGS:

Saturday is the highest-volume day for the entire system with **883,617 trips**.

The insight: our biggest window for conversion is Friday and Saturday. We are catching casual riders when they are most engaged and most likely to benefit from a membership that covers their longer, Friday - Saturday trips.

WEEKLY USAGE PATTERNS BY USER TYPE

Members dominate the workweek; Casual usage surges on weekends



KEY INSIGHTS:

The commuter pattern: annual member usage peaks from **Tuesday to Thursday**, with Thursday reaching a high of **570,439 rides**. This confirms that members primarily use Cyclistic for professional transit.

The weekend surge: casual ridership starts to climb on Friday and peaks sharply on **Saturday (413,676 rides)**.

Converging trends: while members consistently outride casuals, the gap narrows significantly on weekends, especially on Saturday and Sunday.

The bottom line: annual members provide a steady, predictable "floor" for the business during the week, while casual riders provide the "growth ceiling" during the weekends.

EXECUTIVE SUMMARY & STRATEGIC ROADMAP

STRATEGIC RECOMMENDATIONS: CONVERTING LEISURE INTO LOYALTY

KEY TAKEAWAYS:

The behavioral divide: annual members are utility-driven commuters (12 min avg), while casual riders are experience-driven explorers (22 min avg).

The Saturday opportunity: casual engagement peaks on weekends, specifically Saturdays, representing a **413,676-ride window** for targeted conversion.

Data integrity: all findings are based on a verified dataset of 5.5m trips, normalized for DST and time zone accuracy.

WHERE DO WE GO FROM HERE?

1. **Launch "Weekend Warrior" Memberships:** Create a seasonal or weekend-only annual pass specifically priced to appeal to the high-duration Saturday leisure rider.
2. **Targeted Digital Ad Campaigns:** Use geo-fencing at popular recreational stations (parks, waterfronts) on weekends to deliver "Switch & Save" notifications to casual riders.
3. **Value-Based Marketing:** Shift the narrative from "commuting" to "cost-savings." Highlight that for riders averaging 22 minutes, an annual membership pays for itself in just 3 weekend trips per month.

THANK YOU

Olga Chiotoglou